EE115 Lab 6: Second-Order PLL Response

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1 Objective

Compute and plot the phase error response of the second-order PLL, then discuss how the parameters K (loop gain) and a (lead term in $G(s) = 1 + a/s$) affect peak error, speed of convergence, and oscillation frequency. The lab focuses on the underdamped case $4Ka - K^2 > 0$.

2 Theory

For the unit step input to the phase detector, the derived time-domain phase error is

$$\theta_e(t) = \frac{4\pi k_f}{\sqrt{4Ka - K^2}}\, e^{-\frac{K}{2}t}\, \sin\!\big(\sqrt{4Ka - K^2}\, t\big)\, u(t), \qquad 4Ka - K^2 > 0. \tag{1}$$

This is the oscillatory (underdamped) response of the second-order characteristic $s^2 + Ks + Ka = 0$. Using the standard second-order form,

$$\omega_n = \sqrt{Ka}, \qquad \zeta = \frac{K}{2\sqrt{Ka}},$$

so increasing K raises damping and decay rate, while increasing a raises the natural/oscillation frequency.

3 Parameter Choices

All selected pairs satisfy $4Ka - K^2 > 0$ and let us vary one parameter at a time:

- **Vary K (fix $a = 1$):** $K \in \{0.5,\ 1,\ 1.5\}$.
- **Vary a (fix $K = 1$):** $a \in \{0.5,\ 1,\ 2\}$.

Table 1: Validity check for selected parameter pairs ($4Ka - K^2 > 0$).

K	a	$4Ka - K^2$	Note
0.5	1	1.75	Underdamped
1.0	1	3.00	Underdamped
1.5	1	3.75	Underdamped
1.0	0.5	1.00	Underdamped
1.0	2.0	7.00	Underdamped

4 MATLAB Implementation

The script below (also saved as `lab6_plots.m`) follows (1), skips invalid parameter sets, and produces two figures: one where K varies, one where a varies. Replace `kf` with the value specified in your lab handout.

Listing 1: Phase error for multiple (K, a) pairs.

```
clear; clc; close all;

kf = 1;                % replace with lab value if different
t_end = 5;             % extend if the response has not settled
N = 2000;
t = linspace(0, t_end, N);

params_K = [0.5 1; 1 1; 1.5 1]; % vary K, fixed a=1
params_a = [1 0.5; 1 1; 1 2];  % vary a, fixed K=1

make_plot = @(params, titleText, fname) ...
  plot_family(params, t, kf, titleText, fname);

make_plot(params_K, 'Theta_e(t) with a=1, varying K', 'theta_e_vary_K');
make_plot(params_a, 'Theta_e(t) with K=1, varying a', 'theta_e_vary_a');

function plot_family(params, t, kf, titleText, fname)
  figure; hold on; grid on;
  for i = 1:size(params,1)
    K = params(i,1);
    a = params(i,2);
    disc = 4*K*a - K^2;
    if disc <= 0
      warning('Skipping K=%.3g, a=%.3g (4Ka-K^2 <= 0)', K, a);
      continue;
    end
    omega_d = sqrt(disc);
    theta_e = 4*pi*kf/omega_d .* exp(-0.5*K*t) .* sin(omega_d*t);
    plot(t, theta_e, 'DisplayName', sprintf('K=%.2f, a=%.2f', K, a));

    % Optional quick metrics
    peak_val = max(theta_e);
    within = abs(theta_e) <= 0.02*peak_val;
    last_violation = find(~within, 1, 'last');
    if isempty(last_violation)
      t_settle = t(1);
    elseif last_violation < numel(t)
      t_settle = t(last_violation+1);
    else
      t_settle = NaN;
    end
    [pks, locs] = findpeaks(theta_e, t);
    if numel(locs) >= 2
      osc_freq = 1/mean(diff(locs));
    else
      osc_freq = NaN;
    end
    fprintf('K=%.2f, a=%.2f | peak=%.3f, t_settle~=%.3f, f_osc~=%.3f\n', ...
            K, a, peak_val, t_settle, osc_freq);
  end
  xlabel('Time (s)');
  ylabel('\theta_e(t) (rad)');
  title(titleText);
  legend('Location','northeast');
  print(fname, '-dpng', '-r300');
end
```

5 Figures

Plots exported by `lab6_plots.m` (saved as `a=1.png` and `k=1.png`):

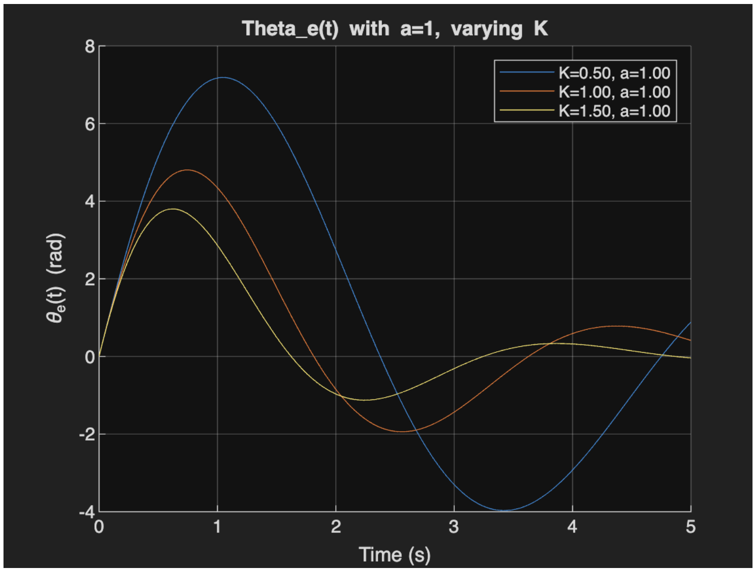

Figure 1: $\theta_e(t)$ with $a = 1$ and varying K.

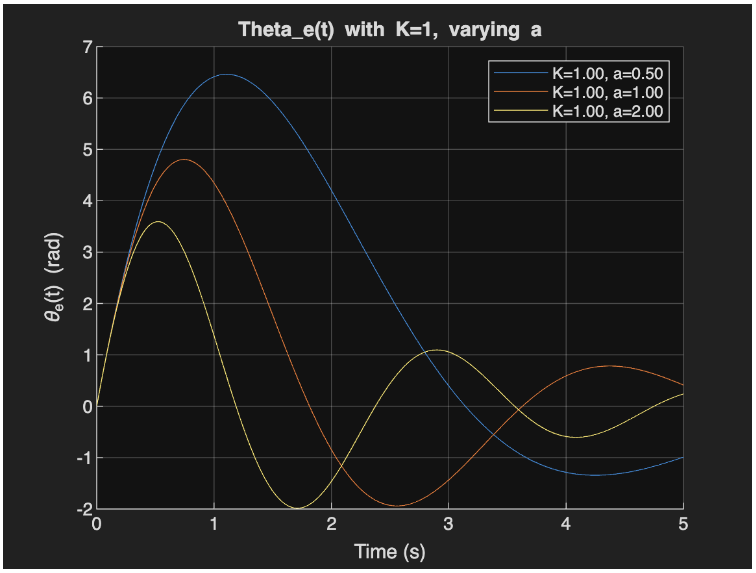

Figure 2: $\theta_e(t)$ with $K = 1$ and varying a.

6 Discussion Points

Use the plotted curves (and printed metrics, if enabled) to answer the lab questions:

- **Peak value:** For the chosen sets, higher K (with a fixed) generally lowers the peak because the higher damping suppresses overshoot. Changing a while holding K fixed shifts the frequency more than the peak amplitude.
- **Speed to converge:** The decay envelope is $e^{-Kt/2}$, so larger K visibly shortens settling time. Varying a at fixed K has minor effect on decay rate; the exponent is unchanged.
- **Oscillation frequency:** $\omega_d = \sqrt{4Ka - K^2}$; increasing a increases ω_d (tighter spacing between zero crossings and peaks). Increasing K slightly reduces ω_d for constant a because of the $-K^2$ term.
- **Stability boundary:** If $4Ka - K^2 \leq 0$, the system loses its oscillatory response (critically damped or unstable). All plotted cases stay in the underdamped region.